

INTERPRETATION OF SEISMIC CROSS-HOLE MEASUREMENTS WITH THE COMBINATION OF TWO METHODS

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The cross-hole survey is a basic technique for the estimation of the subsoil elastic moduli. Although the norms establish a close separation between the boreholes to carry out these surveys, there is no security that all the first seismic arrivals detected by the geophones in one of the holes correspond to direct waves, so not all the calculated velocities are real. It is necessary to correct the deviations from the direct wave time due to refractions in the interfaces or geological contacts. In this work, a combination of two computational methods is used for the interpretation of cross-hole measurements: the first method calculates a horizontal velocity model from the first seismic arrivals curve; the second method calculates the theoretical first seismic arrivals from the stratified subsoil model in order to compare the traveltimes at each depth and improve the subsoil model. The application of the two methods to a cross-hole survey is presented.

1. INTRODUCTION

In geotechnical studies the estimation of dynamic elastic moduli of the subsoil through its seismic velocities is very important for the design of foundations and structures in civil engineering projects. These constants are indicative of its deformability and mechanical resistance. The shear modulus is fundamental for the study of the soil response to seismic or man made vibrations. The shear modulus also allows to understand the soil behavior because it is indicative of its structure. Programs like SHAKE estimate the soil seismic response for a plane layered model (thickness and shear velocity of each layer).

Among the geophysical methods used to obtain the in-situ seismic velocities: refraction, up-hole and cross-hole, the last one gives the most detailed form. The cross-hole method investigates the zone between two close boreholes (generally from 5 to 10 m), measuring the travel time of the compressional (P) and shear (S) waves from a source located in one hole to a geophone located in another hole. In order to obtain a velocity model which is closer to the real subsoil velocities, it is necessary to correct the observed apparent velocities due to refractions in the near layer interfaces.

2. METHODOLOGY

Originally the two programs developed for the interpretation of cross-hole measurements started from a depth model in which the user selected up to 5 depths or interfaces from the apparent velocity curve which is calculated by a first program. However, due to the difficulty to select the depths of real

interfaces (without considering those which can be known from lithology columns), it was decided to fix the layer thickness to 1 m in the model since the dynamic moduli in cross-hole surveys are normally given each meter.

The first program (VELCAP) calculates layer velocities for a given model of up to 14 layers and a half-space. The depths of measurements and the arrival times of P or S waves are the data for the program to calculate the apparent P or S velocities. The user must give the depth of the first layer and the thickness of the layers below if it is constant, or the depth of each layer if the thickness is variable. The program carries out a weighted average of apparent velocities which belong to the same layer and an initial velocity model is calculated. Then, from this model the travel times of the direct and refracted waves in the upper and lower layer are calculated (in the case these are generated) and the minimum is chosen. The possibility of a second refraction is not taken into account by the program, however the close distances used in the cross-hole method make this possibility uncommon. The minimum travel time of the velocity model is compared with the observed travel time and a fix error is evaluated according to the following expression:

$$\text{Err}(\%) = 100 * (\sum_i (T_{mi} - T_{oi}) / T_{oi}) / N_p$$

Where T_{mi} = i-th minimum time

T_{oi} = i-th observed time

N_p = number of points in the i-th layer

The program shows the initial layer velocities calculated from the apparent velocities and the percentage error between the model and observed times for each layer. The user can change the velocity value of one or more layers and recalculate the percentage error until the errors are considered acceptable or can not be reduced significantly, and so a final model is obtained.

The second program (TRECH) prints out the final velocity model and the superposed curves of calculated and observed minimum travel times, in order to compare them and to be sure that the model can reproduce the observations to an acceptable accuracy.

3. APPLICATION EXAMPLE

The cross-hole data of a survey carried out in Villa de Reyes, San Luis Potosi, México (Aranda, 1994) for a power plant site, were used with the two programs described above. In this site there is a shallow layer which is composed of pyroclastic cemented sand, 1 to 2 m thick, which overlies tuff layers with less mechanical strength. In the tuff layers clay content increases while the sand decreases and at 7 m depth the clay predominates. The travel times of P and S waves were measured each 50 cm using two 15 m long boreholes which are separated 4 m each other. A gamma-gamma ray log was run in one of the boreholes to estimate bulk density of the subsoil materials. Caliper and deviation logs were also run in both boreholes for hole-diameter and source-geophone distance corrections, respectively.

Figure 1 shows the minimum travel times of P and S waves in the cross-hole survey. The "I" capital letter which is printed at some depths indicates the velocity inversions with depth, which indicates low velocity layers between higher velocity layers.

Figure 2 shows the output of VELCAP program for an initial P-wave depth model of 14 layers and a half-space. The thickness of all layers is 1 m, except for layer 1 which is 1.2 m thick. The program calculates an initial velocity for each layer using a weighted average. The layer velocities are listed from left to right in lines with five elements. An error (in parenthesis) is also calculated for each layer, by comparison between the estimated travel times (minimum travel times) and the observed travel times; a total error for the whole model is also indicated. After some trials to reduce the adjustment error in the layers the user stops the program and the last velocity model is considered as the final model.

Figure 3 shows the output of the program for the same depth model., but taking into account the S-wave data. During the running test of the program it was observed that the most advisable was to change the velocity of a single layer at a time, in order perceive easily if the error reduces or not, since the reduction of the error in one layer can induce a considerable error increase in one or both adjacent layers.

The final P and S-wave layered models, each one at a time, were the input for the TRECH program, which compares graphically the minimum travel times for the final velocity model and the observed travel times (Fig. 4). It can be observed that the theoretical P and S-wave travel times approach the observed travel times in the majority of the curve. However, in some portions of the curve the similarity in form is poor, the corresponding layer error is big, and it is impossible to reduce significantly the error with any velocity change. This is the case for the 5th and 6th layers in the P-wave velocity model and for the 5th, 6th and 8th layers in the S-wave velocity model. The analysis of the observed travel times reveals that this situation occurs when the consecutive travel times are very similar or equal, but are artificially separated by a layer interface; logically these values should be included in the same layer.

In order to consider the possibility mentioned above, both programs were modified to accept interfaces at any depth and layers with a single travel time, so that the theoretical travel times can fix better the observed travel times. In this way, the layered P and S velocity models could be improved by changing some interface depths and allowing layers with a single point data. Figure 5 shows the TRECH outputs after this changes. The P-wave model was modified first, to obtain the final model whose corresponding travel times show a better similarity with the observed times than the previous model. Accordingly, the same interface depths were considered for the S-wave data but it was not possible to reach such a good fix as with the P-wave data.

Finally the Young and shear moduli were calculated with the final P and S-velocity model which is composed by 15 layers (fig. 6), and superposed with the moduli calculated with the apparent seismic velocities each 0.5 m (point values). Although the moduli for the layered model are not very different from the point values, some important differences can be appreciated between both calculations. The Young modulus for the layered model reaches higher and lower values than the modulus calculated with the apparent velocities; in the shear modulus case, the situation is analogous but the layer and

point values are closer each other. On the other hand, the model clearly shows the presence of some low velocity layers in the subsoil which is emphasized by the hard pyroclastic layer near the surface.

4: CONCLUSIONS

The methods described in this paper and translated in two computer programs to improve the interpretation of cross-hole seismic measurements, are based on calculation of minimum travel times and estimation of a layered model by trial and error.

The semi-automated interpretation procedure proposed needs the user criteria to modify a model with equal thickness layers to a model with variable thickness layers, which in the program are limited to 15. This may give good results for the P times but not for the S times or vice versa, and this can increase the time needed for the interpretation. It would be highly advisable to find a procedure to prove automatically several models to speed up the interpretation, however the non-linear nature of the problem makes difficult to introduce an iterative procedure in the programming. The user decides which is the accuracy necessary in the model to approach the observed time values; on the other hand, the calculations are fast.

The calculated velocity model and the derived layer moduli provide important information about the mechanical resistance of the subsoil. The presence and approximated thickness of low velocity layers, which can constitute a risk for the foundation of civil engineering projects, is specially enhanced which the procedure described.

REFERENCES

Aranda López R., 1994.--Estudio geosísmico de cross-hole en la C.T. Villa de Reyes, S.L.P., C.F.E., México, internal report, unpublished.

I = velocity inversion				I = velocity inversion			
P-WAVE TIMES		S-WAVE TIMES		P-WAVE TIMES		S-WAVE TIMES	
Depth.		Velocity		Depth		Velocity	
.5 +	*	2259.9		.5 +	*	1000.0	
1.0 +		2702.7	*	1.0 +		* 1687.8	
1.5 +		* 3007.5		1.5 + I		* 1619.4	
2.0 + I		2381.0	*	2.0 + I		1159.4	*
2.5 + I		2000.0	*	2.5 + I		1000.0	*
3.0 + I	*	1403.5		3.0 + I	*	753.3	
3.5 +		1754.4	*	3.5 + I	*	676.8	
4.0 +		1754.4	*	4.0 + I	*	627.0	
4.5 + I	*	1550.4		4.5 +	*	647.2	
5.0 + I	*	1201.2		5.0 + I	*	559.4	
5.5 +	*	1250.0		5.5 + I	*	544.2	
6.0 +		1652.9	*	6.0 +	*	764.8	
6.5 + I	*	1351.4		6.5 + I	*	588.2	
7.0 +	*	1351.4		7.0 +	*	603.3	
7.5 + I	*	1230.8		7.5 +	*	632.9	
8.0 +	*	1351.4		8.0 + I	*	603.3	
8.5 +	*	1550.4		8.5 +	*	676.8	
9.0 +	*	1600.0		9.0 +	*	809.7	
9.5 + I	*	1562.5		9.5 + I	*	706.7	
10.0 + I	*	1374.6		10.0 + I	*	618.2	
10.5 +	*	1403.5		10.5 + I	*	608.8	
11.0 + I	*	1250.0		11.0 +	*	735.3	
11.5 +	*	1257.9		11.5 + I	*	723.3	
12.0 +	*	1503.8		12.0 +	*	809.7	
12.5 +	*	1652.9		12.5 +	*	898.9	
13.0 + I	*	1550.4		13.0 + I	*	824.7	
13.5 + I	*	1503.8		13.5 + I	*	809.7	
14.0 + I	*	1454.5		14.0 + I	*	750.5	
14.5 + I	*	1351.4		14.5 + I	*	723.3	

Fig. 1 Observed minimum travel times of P and S-waves

Filename:LOV3.PSA
 Data file is TP1.DAT
 Distance between holes is 4.0 m
 VILLA DE REYES, P-WAVE MODEL

INITIAL DEPTH MODEL (meters)

1.2
 2.2
 3.2
 4.2
 5.2
 6.2
 7.2
 8.2
 9.2
 10.2
 11.2
 12.2
 13.2
 14.2

VELOCITIES AND (ERRORS IN %) (1,2,3,4,.....)

2357.(4.24) 2744.(11.42) 1749.(19.02) 1754.(.00) 1404.(8.06)
 1419.(14.19) 1351.(.20) 1281.(6.12) 1571.(1.58) 1484.(6.29)
 1339.(4.01) 1381.(4.14) 1610.(3.18) 1483.(1.30) 1351.(2.70)

Total error (%)= 5.870

After 14 trials

VELOCITIES AND (ERRORS IN %) (1,2,3,4,.....)

2357.(3.61) 3010.(10.49) 1000.(6.86) 1754.(.00) 1250.(3.19)
 1300.(14.82) 1351.(.00) 1200.(4.97) 1600.(1.55) 1400.(3.79)
 1404.(5.50) 1250.(2.27) 1650.(3.11) 1483.(2.06) 1351.(2.70)

Total error (%)= 4.384

VILLA DE REYES, P-WAVE MODEL

FINAL MODEL
 DEPTH VELOCITY Error(%)

1.2	2357.	3.61
2.2	3010.	10.49
3.2	1000.	6.86
4.2	1754.	.00
5.2	1250.	3.19
6.2	1300.	14.82
7.2	1351.	.00
8.2	1200.	4.97
9.2	1600.	1.55
10.2	1400.	3.79
11.2	1404.	5.50
12.2	1250.	2.27
13.2	1650.	3.11
14.2	1483.	2.06
.	1351.	2.70

Fig. 2 Output of VELCAP program for the P-wave travel times.

Filename:LOV3.SSA
 Data file is TS1.DAT
 Distance between holes is 4.0 m
 VILLA DE REYES, S-WAVE MODEL

INITIAL DEPTH MODEL (meters)

1.2
 2.2
 3.2
 4.2
 5.2
 6.2
 7.2
 8.2
 9.2
 10.2
 11.2
 12.2
 13.2
 14.2

VELOCITIES AND (ERRORS IN %) (1,2,3,4,.....)

1151.(20.07) 1426.(16.13) 896.(18.95) 656.(9.18) 610.(7.09)
 637.(17.32) 595.(1.23) 620.(7.24) 733.(9.07) 670.(5.28)
 662.(5.14) 760.(3.16) 868.(4.27) 785.(2.23) 723.(2.02)

Total error (%)= 8.783

After 17 trials

VELOCITIES AND (ERRORS IN %) (1,2,3,4,.....)

1000.(13.09) 1687.(17.64) 500.(7.14) 656.(3.80) 640.(8.84)
 700.(15.76) 500.(3.16) 633.(8.97) 790.(8.41) 670.(4.90)
 640.(3.93) 760.(3.16) 868.(4.27) 785.(2.23) 723.(2.02)

Total error (%)= 7.331

VILLA DE REYES, S-WAVE MODEL

FINAL MODEL

DEPTH VELOCITY Error(%)

1.2	1000.	13.09
2.2	1687.	17.64
3.2	500.	7.14
4.2	656.	3.80
5.2	640.	8.84
6.2	700.	15.76
7.2	500.	3.16
8.2	633.	8.97
9.2	790.	8.41
10.2	670.	4.90
11.2	640.	3.93
12.2	760.	3.16
13.2	868.	4.27
14.2	785.	2.23
.	723.	2.02

Fig. 3 Output of VELCAP program for the S-wave travel times

VILLA DE REYES, P-WAVE

MODEL		
LAYER	DEPTH (m)	VELOCITY (m/s)
1	1.20	2360.
2	2.20	3010.
3	3.20	1000.
4	4.20	1754.
5	5.20	1250.
6	6.20	1300.
7	7.20	1350.
8	8.20	1200.
9	9.20	1600.
10	10.20	1400.
11	11.20	1405.
12	12.20	1250.
13	13.20	1650.
14	14.20	1480.
15	.	1350.

P-WAVE TIMES

T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)		T.estimated(*)		T.estimated(#)			
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VILLA DE REYES, S-WAVE MODEL

MODEL		
LAYER	DEPTH (m)	VELOCITY (m/s)
1	1.20	1000.
2	2.20	1687.
3	3.20	500.
4	4.20	656.
5	5.20	640.
6	6.20	700.
7	7.20	500.
8	8.20	633.
9	9.20	790.
10	10.20	670.
11	11.20	640.
12	12.20	760.
13	13.20	870.
14	14.20	785.
15	.	720.

S-WAVE TIMES

T.estimated(*)		T.observed(#)	T.Obs ap.V I Layer		
Depth (m)	T.Cal (ms)		(ms)	(m/s)	
.5	3.50+	* #	4.00	1000.	1
1.0	2.69+	# *	2.37	1688.	1
1.5	2.37+	#	2.47	1619.	2
2.0	2.37+	* #	3.45	1159.	2
2.5	3.52+	* #	4.00	1000.	3
3.0	5.43+	#*	5.31	753.	3
3.5	6.10+	# *	5.91	677.	4
4.0	6.10+	* #	6.38	627.	4
4.5	6.25+	#*	6.18	647.	5
5.0	5.97+	* #	7.15	559.	5
5.5	5.71+	*	7.35	544.	6
6.0	5.71+	# *	5.23	765.	6
6.5	6.55+	* #	6.80	588.	7
7.0	6.81+	#*	6.63	603.	7
7.5	6.32+	#	6.32	633.	8
8.0	5.44+	* #	6.63	603.	8
8.5	5.06+	* #	5.91	677.	9
9.0	5.06+	#*	4.94	810.	9
9.5	5.54+	*#	5.66	707.	10
10.0	5.97+	* #	6.47	618.	10
10.5	6.25+	* #	6.57	609.	11
11.0	5.60+	#*	5.44	735.	11
11.5	5.26+	* #	5.53	723.	12
12.0	4.85+	*#	4.94	810.	12
12.5	4.60+	#*	4.45	899.	13
13.0	4.60+	* #	4.85	825.	13
13.5	4.93+	#	4.94	810.	14
14.0	5.10+	* #	5.33	750.	14
14.5	5.43+	#	5.53	723.	15

Fig. 4 Output of TRECH program : a) for P-wave travel times and model
b) for S-wave travel times and model

VILLA DE REYES. P-WAVE NEW MODEL

MODEL		
LAYER	DEPTH (m)	VELOCITY (m/s)
1	1.20	2360.
2	2.20	3010.
3	3.20	1000.
4	4.20	1754.
5	4.70	1550.
6	5.70	700.
7	6.20	1653.
8	7.20	1350.
9	8.20	1225.
10	9.20	1570.
11	10.70	1390.
12	11.70	1100.
13	12.70	1566.
14	13.70	1530.
15	.	1430.

P-WAVE TRAVELTIMES

Depth (m)	T.Cal (ms)	T.estimated(*)	T.observed(##)	T.Obs ap.V (ms) (m/s)	Layer
.5	1.69+		*#	1.77 2260.	1
1.0	1.43+		#	1.48 2703.	1
1.5	1.33+		#	1.33 3008.	2
2.0	1.33+	*	#	1.68 2381.	2
2.5	1.89+		* #	2.00 2000.	3
3.0	2.61+		* #	2.85 1404.	3
3.5	2.28+		#	2.28 1754.	4
4.0	2.28+		#	2.28 1754.	4
4.5	2.46+		* #	2.58 1550.	5
5.0	3.35+			# 3.33 1201.	6
5.5	2.94+		* #	3.20 1250.	6
6.0	2.42+		#	2.42 1653.	7
6.5	2.68+		* #	2.96 1351.	8
7.0	2.96+		#	2.96 1351.	8
7.5	3.17+			*# 3.25 1231.	9
8.0	2.75+		* #	2.96 1351.	9
8.5	2.55+		#	2.58 1550.	10
9.0	2.55+		#*	2.50 1600.	10
9.5	2.75+		# *	2.56 1563.	11
10.0	2.88+		*#	2.91 1375.	11
10.5	2.88+		#	2.85 1404.	11
11.0	3.21+		#	3.20 1250.	12
11.5	2.81+		* #	3.18 1258.	12
12.0	2.55+		* #	2.66 1504.	13
12.5	2.55+		# *	2.42 1653.	13
13.0	2.61+		#*	2.58 1550.	14
13.5	2.61+		*#	2.66 1504.	14
14.0	2.76+		#	2.75 1455.	15
14.5	2.80+		* #	2.96 1351.	15

VILLA DE REYES, S-WAVE NEW MODEL

MODEL		
LAYER	DEPTH (m)	VELOCITY (m/s)
1	1.20	1000.
2	2.20	1687.
3	3.20	500.
4	4.20	656.
5	4.70	650.
6	5.70	450.
7	6.20	765.
8	7.20	595.
9	8.20	620.
10	9.20	733.
11	10.70	625.
12	11.70	730.
13	12.70	850.
14	13.70	820.
15	.	744.

S-WAVE TRAVELTIMES

T.estimated(*)		T.observed(#)		T.Obs ap.V		Layer
Depth (m)	T.Cal (ms)			(ms)	(m/s)	
.5	3.50+	*	#	4.00	1000.	1
1.0	2.69+	#	*	2.37	1688.	1
1.5	2.37+	#		2.47	1619.	2
2.0	2.37+	*	#	3.45	1159.	2
2.5	3.52+	*	#	4.00	1000.	3
3.0	5.43+		#*	5.31	753.	3
3.5	6.10+		# *	5.91	677.	4
4.0	6.10+		* #	6.38	627.	4
4.5	6.15+		#	6.18	647.	5
5.0	7.12+		*#	7.15	559.	6
5.5	5.95+		*	# 7.35	544.	6
6.0	5.23+		#	5.23	765.	7
6.5	5.86+		* #	6.80	588.	8
7.0	6.64+		#	6.63	603.	8
7.5	6.45+		#*	6.32	633.	9
8.0	5.80+		* #	6.63	603.	9
8.5	5.46+		* #	5.91	677.	10
9.0	5.46+		# *	4.94	810.	10
9.5	5.96+		# *	5.66	707.	11
10.0	6.40+		#	6.47	618.	11
10.5	5.81+		* #	6.57	609.	11
11.0	5.48+		#	5.44	735.	12
11.5	4.99+		* #	5.53	723.	12
12.0	4.71+		* #	4.94	810.	13
12.5	4.71+		# *	4.45	899.	13
13.0	4.88+		#*	4.85	825.	14
13.5	4.88+		#	4.94	810.	14
14.0	5.22+		*#	5.33	750.	15
14.5	5.38+		*#	5.53	723.	15

Fig. 5 Output of TRECH program for the improved depth model :
a) for the P-wave travel times b) for the S-wave travel times